Introducción a Tecnología CMOS

Complementary Metal Oxide Semiconductors

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Universidad Nacional de Colombia

Agenda

MOSFET Transistors

Body Effect

CMOS Technology

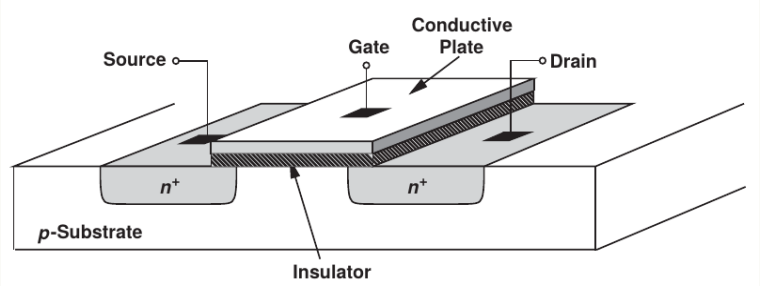
Symbols

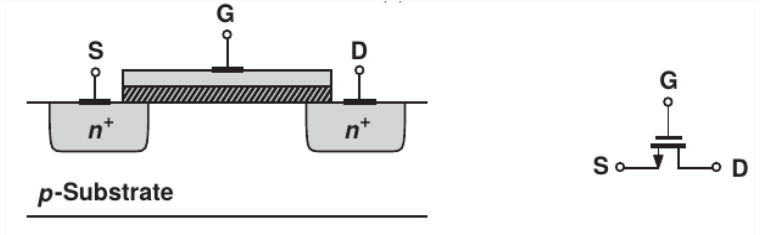
NMOS Inverter

CMOS Inverter

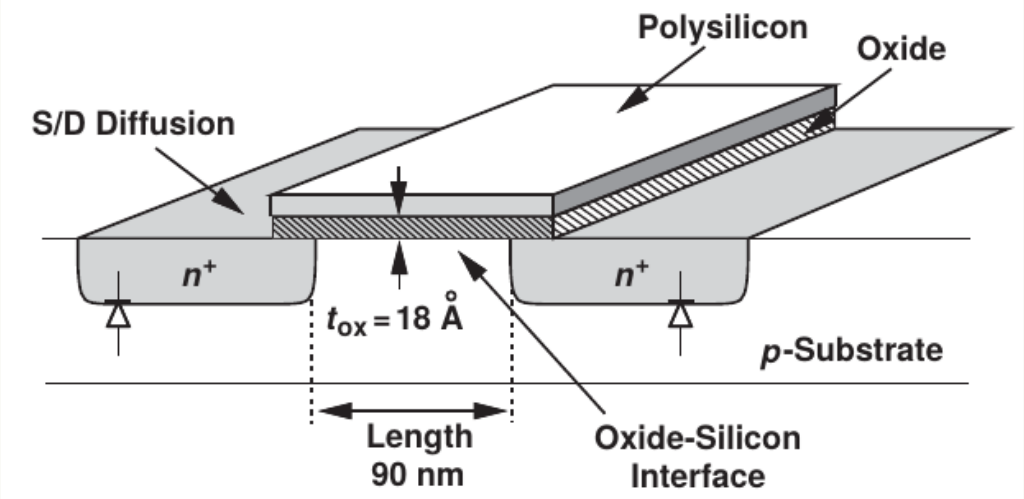
MOSFET Transistors

¿Qué es un Transistor MOSFET?

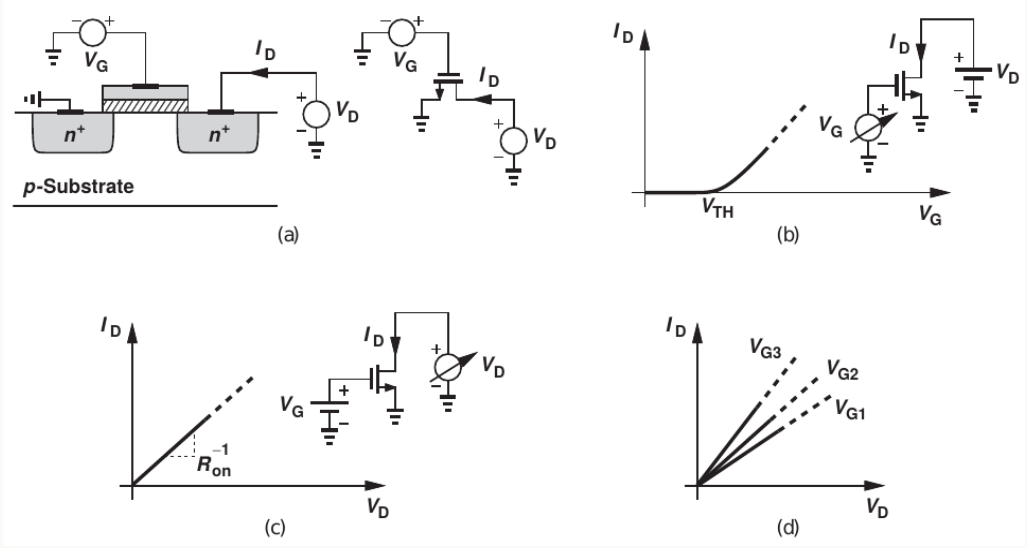




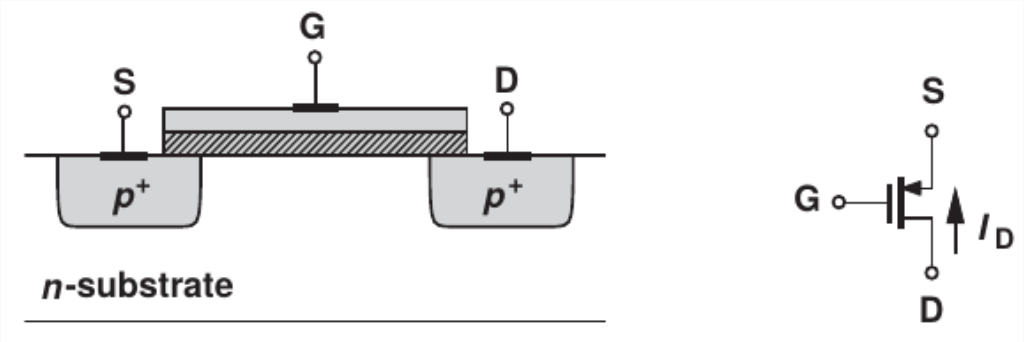
A little bit more detail...



Some important behavior of this transistor...

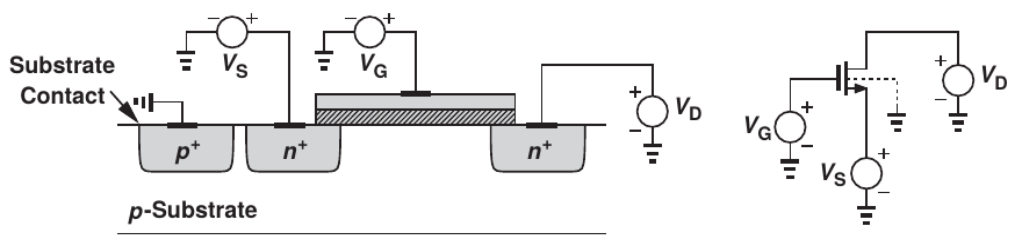


PMOS Transistor



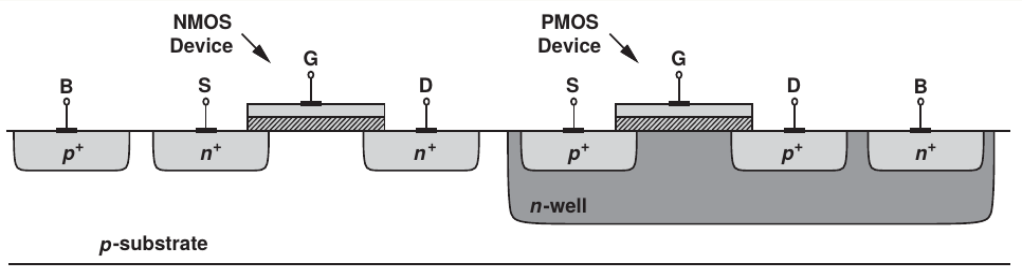
Body Effect

MOSFET Body Effect



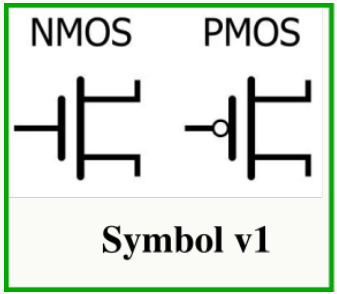
CMOS Technology

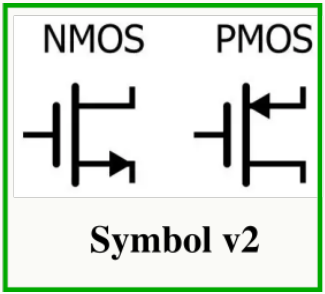
CMOS Technology

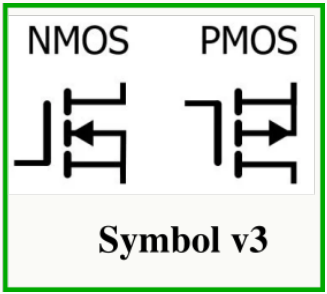


Symbols

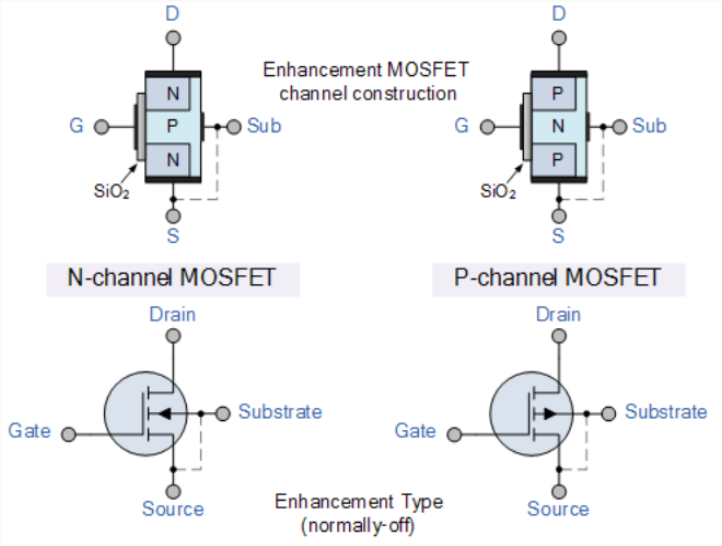
MOSFET Transistors Symbols





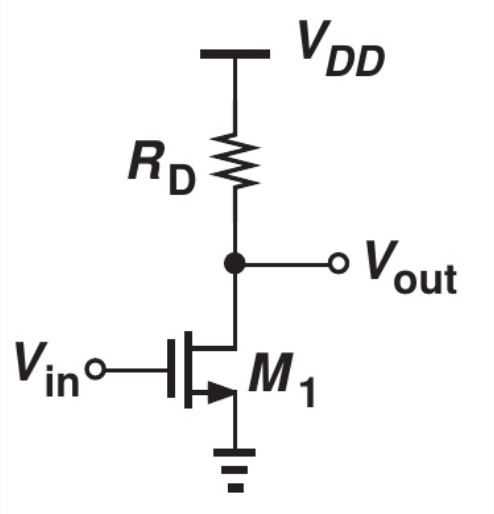


MOSFET Transistors Symbols



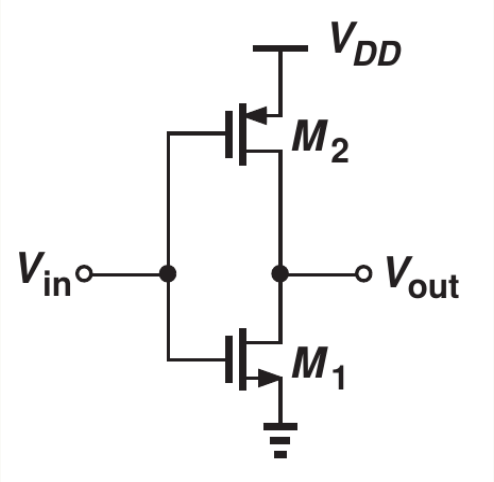
NMOS Inverter

NMOS Inverter

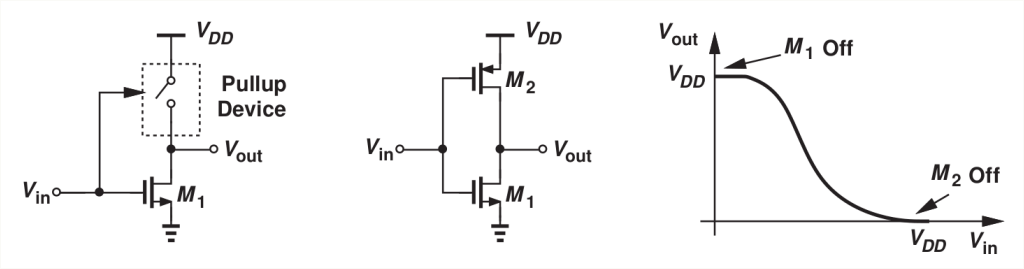


CMOS Inverter

CMOS Inverter

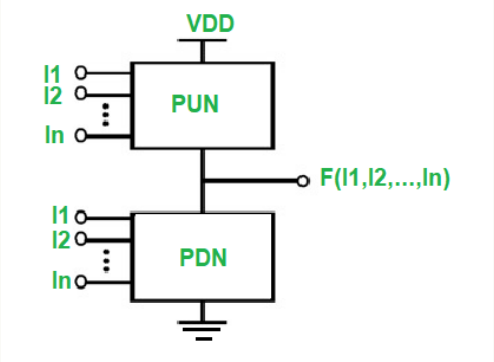


CMOS Inverter — Voltage Transfer Characteristic



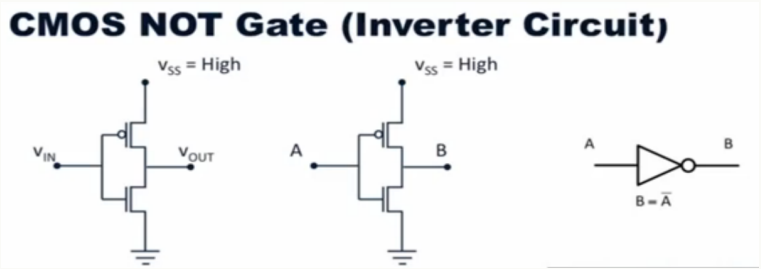
- Explica *por qué* el circuito se comporta como inversor: transición abrupta, oscilación completa (rail-to-rail)

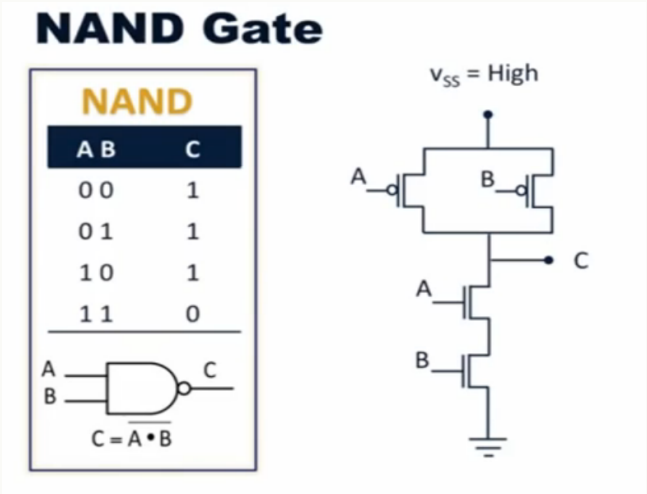
Construcción General de Compuertas CMOS

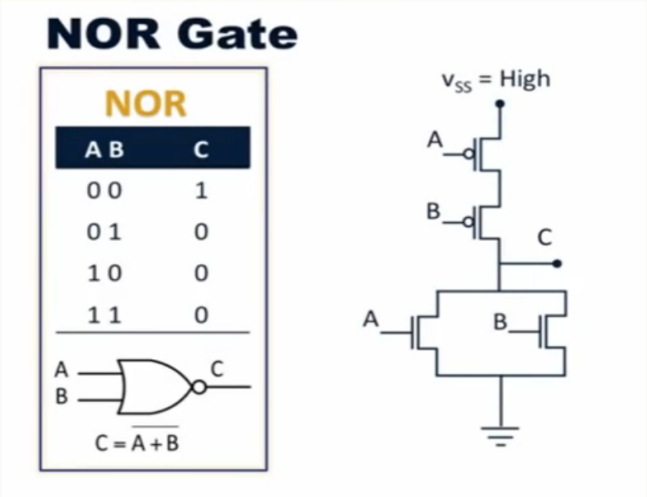


- **PDN** (red de bajada, NMOS): conduce cuando la salida debe ser `0`
- **PUN** (red de subida, PMOS): conduce cuando la salida debe ser `1`
- Redes complementarias $\Rightarrow$ sin camino DC de V_{DD} a GND en estado estable $\Rightarrow$ potencia estática casi nula. NAND y NOR son *instancias* de esta regla general

CMOS Inverter




NAND Gate
NAND
A B C
0 0 1
0 1 1
1 0 1
1 1 0
A
B
C
C = A • B
v_{SS} = High
A
B
C
A
B


NOR Gate
NOR
A B C
0 0 1
0 1 0
1 0 0
1 1 0
A
B
C
C = A + B
v_{SS} = High
A
B
C
A
B

Créditos

Las imágenes presentadas en las diapositivas 10, 15, 16 y 17 de esta presentación fueron tomadas de páginas de internet de acceso libre.

Las imágenes presentadas en las diapositivas 3, 4, 5, 6, 7 y 8 fueron tomadas del capítulo 6 y las imágenes de las diapositivas 11, 12 y 13, fueron tomadas del capítulo 16 del libro *"Fundamentals of Microelectronics"*;

Razavi, Behzad. "Fundamentals of Microelectronics", 2nd Edition. WILEY. 2014

La imágen presentada en las diapositiva 14 fue tomada de la página de internet *www.geeksforgeeks.org*